

The Pleistocene-Holocene Transition in Tagua-Tagua, central Chile

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Abstract

Paleohydrological reconstructions from the Chilean Altiplano document abrupt moisture fluctuations during the last millennia. Although the end of the mid Holocene aridity and the onset of more humid conditions between 6–4 ka has been identified in numerous regional marine and terrestrial sites, the timing of late Holocene dry and humid phases shows large regional variability. Laguna Miscanti and Laguna Miñiques (23° 43'S – 67° 46'W, 4140 m asl) are topographically closed, but connected by surface outflow from Miscanti. Sedimentological and geochemical indicators from two new cores show large facies changes, i.e. higher carbonate and evaporite deposition during more arid periods and increased organic productivity (both algal and macrophyte) during more humid phases. As in most Andean lakes located in volcanic settings, large ¹⁴C reservoir effects occur complicating ¹⁴C dating, so the age models include ²¹⁰Pb and U/Th dating. In spite of dating uncertainties, both lakes show similar patterns during the last millennium. A humid phase in Laguna Miscanti prior to ca 1200 CE is coherent with rodent middens and geomorphological features indicative of a major pluvial/recharge event at lower altitudes (Atacama Desert) during the Medieval Climate Anomaly (ca 800 - 1300 CE). The LIA (1300 – 1850 CE) is characterized by several arid/humid cycles and the last century by a productivity increase. The hydrological changes observed during the last millennium illustrate the complex dynamics of recent climate evolution over the high altitude Andean plateau. Discrepancies between the northern and southern Altiplano records and with intermediate latitudes (Central Chile) records may reflect contrasting responses to external forcing (Westerlies versus South American Monsoon dynamics) along different climatic zones.

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18 **1. Introduction**

19 **2. Study site and regional setting**

20 **3. Materials and methods**

21 *3.1. Sedimentary and geochemical analysis*

22 In Diciembre 2013, we recovered 5 cores sediment at several sites around to
 23 Tagua Tagua 3 site using sing a hammer-modified UWITEC gravity corer and a
 24 UWITEC platform (Fig. 1 and Table S1). The cores were transported to the
 25 IACT-CSIC (Granada) and IPE-CSIC (Zaragoza-Jaca) laboratories in Spain for
 26 image acquisition and measurement of sediment physical properties, density and
 27 Magnetic Susceptibility (MS [$\times 10^{-8}$ (m^3/kg))]

28 at 1 cm resolution, with a Geotek Multi-Sensor Core Logger (MSCL). Ele-
 29 mental geochemical analyses were performed every 1 cm for total carbon (TC),
 30 total inorganic carbon (TIC), total organic carbon (TOC=TC-TIC) and total
 31 sulfur (TS) in a LECO SC 144 DR furnace (IPE-CSIC). Total nitrogen (TN) was
 32 measured in a VARIO MAX CN elemental analyzer (IPE-CSIC). Biogenic opal
 33 (BioSi) was measured using a continuous flow analyzer AutoAnalyser Technicon
 34 II at the Instituto de Investigaciones Marinas (IIM-CSIC, Spain) (Bernárdez et
 35 al., 2005). An AVAATECH X-Ray Fluorescence II core scanner with an Rh x-ray
 36 tube from the University of Barcelona was used to obtain x-ray fluorescence
 37 (XRF) logs from the composite sequence at 4 mm resolution. Thus, differents
 38 ratios have been calculated here to describe changes in the carbonate produc-
 39 tivity (Ca/K and Ca/Ti in Miscanti and Miñique, respectively), local runoff
 40 (Rb/Sr), grain size (Zr/Rb), evaporative concentration of gypsum (Ca/[Rb/Sr]
 41 and Ca/Ti in Miscanti and Miñique, respectively) and organic productivity
 42 (Br/K and Br/Ti in Miscanti and Miñique, respectively) (Carrevedo et al., 2015;
 43 Frugone-Álvarez et al., 2020; Frugone-Álvarez et al., 2017; Moreno et al., 2007;
 44 Naeher et al., 2013; Wersin et al., 1991). Sedimentary facies were described
 45 based on criteria formulated by (Schnurrenberger et al., 2003) and lacustrine
 46 units have been classified according to the sedimentary structure, MSCL data,
 47 elemental composition (TOC, TIC, TN, TS, TOC/TN, BioSi), XRF data and
 48 waxes composition.

49 *3.2. Chronology*

50 The chronology for the Miscanti (MIS13-2A) and Miñique (MIN13-1F) se-
 51 quence was obtained using ^{210}Pb and ^{137}Cs dating techniques, eight U/Th dates
 52 and, nine ^{14}C AMS dates (see Table 1). Core MIN13-1F-1G was sampled in
 53 the field every 0.5 and 1 cm for ^{210}Pb and ^{137}Cs . The MIS13-2A-1G sequence
 54 was correlated with the ^{210}Pb and ^{137}Cs data from the sequence obtained by
 55 Valero-Garcés et al., 1996. The radiocarbon ages were determined in MIS13-
 56 2A-1G and MIN13-1F-1G cores by AMS at Poznan Radiocarbon Laboratory,
 57 Poland and the DirectAMS laboratories, USA. Waters from the hypolimnion has

also sampled to assess modern ^{14}C reservoir effects (Table 1). The U/Th dating was done at the Massachusetts Institute of Technology (MIT, McGee Lab). For U/Th samples, the cores were visually assessed for material to dating criteria and methods described in (Chen et al., 2020; Lawrence Edwards et al., 1987; Shen et al., 2002). A VG PQ2+ quadrupole ICP-MS and an Agilent 7900 ICP-MS at MIT was using to determined uranium and thorium concentration. Samples with higher U/Th ratios were then identified as materials worth further processing as they are more likely to yield “clean” samples with high $^{230}\text{Th}/^{232}\text{Th}$ ratios. The isotopic compositions of the resulting uranium and thorium fractions were then measured on a Nu Plasma II-ES multi-collector ICP-MS at MIT.

3.3. Data analysis

4. Results

4.1. Cores correlations and sedimentary proxies

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